UNIT-3

Date :

WAVE NATURE OF PARTICLES AND SCHRODINGER EQUATION

Macroscopic word — Classical Mechanics

$v = u + at$

$v^2 - u^2 = 2as$

$s = ut + \frac{1}{2}at^2$

Microscopic world — electron, proton, neutrol.

Dual nature of light

→ In some cases light behaves as wave in other cases it behaves as a particles.

* Wave Nature	* Particles Nature
(i) Interference	(i) Photoelectric effect
(ii) Diffraction	(ii) compton scattering
(iii) Polarization	(iii) Pair Production

de - Broglie Hypothesis

→ According to de Broglie Hypothesis every moving particles having mass 'm' characteristics of both wave and particles.

– The wavelength associated with particles with mass 'm' is

$$\lambda_{de\,brog} = \frac{h}{mv}$$

h = Plank's constant $= 6.634 \times 10^{-34}$ Js

m = Mass of particle

v = Velocity of particles

wave function $\psi(x,t) \rightarrow$ 1-D

$\psi(x,y,t) \rightarrow$ 2-D

$\psi(x,y,z,t) \rightarrow$ 3-D

→ wave function gives us the state of a microscopic particle of mass 'm' at point 'x' at any time 't'.

The wave function of particles having mass 'm' is represented by

$$\psi(x,t) = A e^{i(kx - \omega t)} \quad \text{(1)}$$

$K = \frac{2\pi}{\lambda} \rightarrow$ wave vector

$\omega = 2\pi\nu$ angular frequency

ν = frequency

$\lambda = \frac{h}{mv} = \frac{h}{p}$ p = linear momentum

$E = h\nu$, $\nu = \frac{E}{h}$

$\left[\because \hbar = \frac{h}{2\pi}\right]$

$$K = \frac{2\pi}{\lambda} = \frac{2\pi}{h/p} = \frac{2\pi p}{h} = \frac{p}{h/2\pi} = \frac{p}{\hbar}$$

h = Plank's constant

$\hbar$ = Reduced value of plank's constant.

$$\omega = 2\pi\nu = \frac{2\pi E}{h} = \frac{E}{h/2\pi} = \frac{E}{\hbar}$$

Put these value in eq (1)

Date :

$$\psi(x,t) = A e^{i\left(\frac{p}{\hbar}x - \frac{Et}{\hbar}\right)}$$

$$\boxed{\begin{aligned} \psi(x,t) &= A e^{\frac{i}{\hbar}(px - Et)} \\ \psi(x,t) &= A e^{i(kx - \omega t)} \end{aligned}}$$

Remember.

Uncertanity Relation

Heisenberg uncertainity Relation.

→ According to Uncertainity Relation it is impossible to simultaneously find the position and momentum of a micro-scopic particles with 100% accuracy.

Δx = Error or uncertanity in the measurement of position of particles.

Δp = momentum of particles.

then

$$\boxed{\Delta x \, \Delta p \geq \frac{\hbar}{2}}$$

If $\Delta x = 0$ → error in position = 0

$$\Delta p \geq \frac{\hbar}{2 \times \Delta x}$$

$$\Delta p \geq \frac{1}{0} = \infty$$

If $\Delta p = 0$

$$\Delta x = \frac{h}{\Delta p} = \frac{h}{0} = \infty$$

Born Interpretation

→ According to Born interpretation the probability of existance of a particle at any point 'x' is

$$\boxed{P = \psi^* \psi \big|_{x = x_0} = |\psi|^2}$$

In a region $a \leq x \leq b$, the probability of existance is given by

$$\boxed{P = \int_a^b \psi^* \psi \, dx}$$

$$\psi(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$$

$$\psi^*(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L}$$

(i) Que. find probability at point $x = \frac{L}{2}$

$$P = \psi^* \psi \big|_{x = \frac{L}{2}} = \frac{2}{L} \sin^2 \frac{\pi x}{L} \Big|_{x = \frac{L}{2}}$$

$$\frac{2}{L} \sin^2 \frac{\pi}{\cancel{L}} \cdot \frac{\cancel{L}}{2} = \frac{2}{L} \sin^2 \frac{\pi}{2}$$

$$\boxed{P \text{ at point } x = \frac{L}{2} = \frac{2}{L}}$$

$$P = \int_0^{L/2} \sqrt{\frac{2}{L}} \sin \frac{\pi x}{L} =$$

P.T.O

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Ques. Using Heisenberg Uncertainity Relation Prove that electron cannot exist inside the nucleus.

4 marks

Proof : The Heisenberg Uncertainity Relation

$$\Delta x \; \Delta p \approx \frac{\hbar}{2} \qquad \hbar = \frac{h}{2\pi}$$

$$\Delta x \; \Delta p \approx \frac{h}{4\pi} \quad \text{---} (1)$$

If electron is inside the nucleus

$\Delta x = 10^{-15} m$ = size of nucleus.

$\Delta p = m \Delta v$ $\quad$ ∴ v = velocity of electron

$m = 9.1 \times 10^{-31}$ kg = mass of e^-

Put these values in eq (1)

$$\Delta x \cdot m \Delta v = \frac{h}{4\pi}$$

$$\Delta v = \frac{h}{4\pi m \Delta x}$$

$$= \frac{6.6 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 10^{-15}}$$

$$\Delta v \approx 10^{10} ms^{-1}$$

which is larger than the speed of light is impossible. Hence electron cannot exist inside the nucleus.

i = iota

i^* = complex conjuate = $-i$

Date :

Born Interperetation.

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$\theta = \frac{\pi x}{L}$$

$$\sin^2\frac{\pi x}{L} = \frac{1-\cos\frac{2\pi x}{L}}{2}$$

Put in (1)

$$P = \frac{2}{L}\int_0^{L/2} \frac{1-\cos\frac{2\pi x}{L}}{2}\,dx$$

$$P = \frac{1}{L}\left[\int_0^{L/2} 1.dx - \int_0^{L/2}\cos\frac{2\pi x}{L}\,dx\right]$$

$$= \frac{1}{L}\left[\left[x\right]_0^{L/2} \left[\frac{\sin\frac{2\pi x}{L}}{\frac{2\pi}{L}}\right]_0^{L/2}\right]$$

$$= \frac{1}{L}\left[\left(\frac{L}{2}-0\right) - \frac{L}{2\pi}\left(\sin\frac{2\pi}{L}\cdot\frac{L}{2} - \sin 0\right]\right.$$

$$= \frac{1}{\cancel{L}}\times\frac{\cancel{L}}{2}$$

$$= \cancel{\frac{L}{2}}\quad \boxed{P = \frac{1}{2}}$$

Date :

Expectation value (Average value)

$$\langle f(x)\rangle = \frac{\int_{-\infty}^{\infty} \psi^* f(x)\, \psi\, dx}{\int_{-\infty}^{\infty} \psi^* \psi\, dx}$$

$\langle x\rangle$ = Expectation or average value of position.

$\langle p\rangle$ = Expectation or average value of linear momentum.

Marks

Normalization of wave function

Condition to normalize a wave function is

$$\int_{-\infty}^{\infty} \psi^* \psi\, dx = 1$$

If ψ is normalized

Ques. Consider a ~~wave~~ un normalized wave function

$$\psi(x) = A \sin \frac{\pi x}{L} \quad ; \quad 0 \le x \le L$$

$\therefore$ A = normalized constant

Normalized this wave function.

Soln :- The condition to normalized a wave function

$$\int_{-\infty}^{\infty} \psi^* \psi\, dx = 1$$

$$\psi^* = \psi = A \sin \frac{\pi x}{L}$$

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$$\int_0^L A \sin\frac{\pi x}{L} \cdot A \sin\frac{\pi x}{L}\, dx = 1$$

$$A^2 \int_0^L \sin^2\frac{\pi x}{L}\, dx = 1$$

$$A^2 \int_0^L \frac{1-\cos\frac{2\pi x}{L}}{2}\, dx = 1$$

$$\frac{A^2}{2}\left[x - \frac{L}{2\pi}\sin\frac{2\pi x}{L}\right]_0^L = 1$$

$$\frac{A^2}{2}\left[(L-0) - 0\right] = 1$$

$$A^2\frac{L}{2} = 1$$

$$A^2 = \frac{2}{L}$$

$$A = \sqrt{\frac{2}{L}}$$

Put $A = \sqrt{\frac{2}{L}}$ is given wave function.

Hence, the normalized wavefunction is

$$\boxed{\psi(x) = \sqrt{\frac{2}{L}}\sin\frac{\pi x}{L}}$$

Date :

U.Q.I
6 Marks

Time dependent Schrodinger Equation

Consider a particle of mass 'm' represented by wave function.

$$\psi(x,t) = Ae^{\frac{i}{\hbar}(Et - px)} \quad \text{--- (1)}$$

E = Energy
p = Linear momentum
x = position
t = time

Total energy of particle

$$E = K.E + P.E$$

$$E = \frac{1}{2}mv^2 + V(x)$$

$$E = \frac{1}{2}\frac{m^2v^2}{m} + V(x)$$

$$E = \frac{p^2}{2m} + V(x)$$

Apply wave function ψ both sides

$$E\psi = \frac{p^2}{2m}\psi + V(x)\psi \quad \text{--- (2)}$$

Take derivative of (1) both sides w.r.t 't'

$$\frac{d\psi}{dt} = Ae^{\frac{-i}{\hbar}(Et - px)} \cdot \frac{-iE}{\hbar}$$

$$\frac{d\psi}{dt} = \frac{-iE}{\hbar} Ae^{\frac{-i}{\hbar}(Et - px)}$$

$$\frac{d\psi}{dt} = \frac{-iE}{\hbar}\psi$$

$$\boxed{\frac{\hbar}{-i}\frac{d\psi}{dt} = E\psi} \quad \text{--- (3)}$$

Date :

Take derivative of (1) both sides w.r.t 'x' two times.

$$\frac{d\psi}{dx} = Ae^{\frac{-i}{\hbar}(Et - px)}$$

$$= Ae^{\frac{-i}{\hbar}(Et - px)} \cdot -\frac{i}{\hbar}(-p)$$

$$\frac{d}{dx}\left(\frac{d\psi}{dx}\right) = \frac{A\,ip}{\hbar} \frac{d}{dx}\left[e^{-\frac{i}{\hbar}(Et - px)}\right]$$

$$\frac{d^2\psi}{dx^2} = \frac{ip}{\hbar} A e^{-\frac{i}{\hbar}(Et - px)} \frac{ip}{\hbar}$$

$$\frac{d^2\psi}{dx^2} = \frac{i^2 p^2}{\hbar^2} \underbrace{Ae^{-\frac{i}{\hbar}(Et - px)}}_{=\psi}$$

$$\frac{d^2\psi}{dx^2} = \frac{-p^2\psi}{\hbar^2}$$

$$\boxed{p^2\psi = -\hbar^2 \frac{d^2\psi}{dx^2}} \text{ ------ } (4)$$

put $E\psi = \frac{\hbar}{i}\frac{d\psi}{dt}$ and $p^2\psi = -\hbar^2\frac{d^2\psi}{dx^2}$

from (3) and (4) in eq (2)

$$\frac{\hbar}{-i}\frac{d\psi}{dt} = \frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi$$

multiple and divide by i

$$\frac{i\hbar}{-i^2}\frac{d\psi}{dt} = \frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi$$

Date :

$$\boxed{i\hbar \frac{d\psi}{dt} = \frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi}$$

This is time-depend schrodinger eqn in 1-D

for free particle $V(x) = 0$

$$i\hbar \frac{d\psi}{dt} = \frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2}$$

Time - Independent Schrodinger Eqn.

Time-depend schrodinger equation is

$$\frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x,t) = i\hbar\frac{d\psi}{dt} \quad \text{---- (1)}$$

let the time dependent wave function

$\psi(x,t)$ is written as

$$\psi(x,t) = \phi(x)\, e^{-\frac{iEt}{\hbar}} \quad \text{------- (2)}$$

Take derivative of (2) both sides w.r.t 'x' two times

$$\frac{d\psi}{dx} = \frac{d\phi(x)}{dx} e^{-\frac{iEt}{\hbar}}$$

$$\boxed{\frac{d^2\psi}{dx^2} = \frac{d^2\phi}{dx^2} e^{-\frac{iEt}{\hbar}}} \quad \text{----- (3)}$$

Take derivative of (2) both sides w.r.t 't'

$$\frac{d\psi}{dt} = \phi(x)\, e^{-\frac{iEt}{\hbar}} \cdot \left(-\frac{iE}{\hbar}\right) \quad \text{--- (4)}$$

Date :

Put values of $\frac{d^2\psi}{dx^2}$ and $\frac{d\psi}{dt}$ and ψ

from (3) and (4) in eq (1)

$$-\frac{\hbar^2}{2m}\left(\frac{d^2\phi}{dx^2}\cancel{e^{-\frac{iEt}{\hbar}}}\right) + V(x)\left[\phi(x)\cancel{e^{-\frac{iEt}{\hbar}}}\right] = i\hbar\left[\phi(x)\cancel{e^{-\frac{iEt}{\hbar}}}\left(\frac{-iE}{\hbar}\right)\right]$$

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\phi(x)}{dx^2} + V(x)\phi(x) = E\phi(x)}$$

Time-independent ~~schon~~ schrodinger eqn in 1-D for particle $\boxed{V(x)=0}$

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\phi}{dx^2} = E\phi}$$

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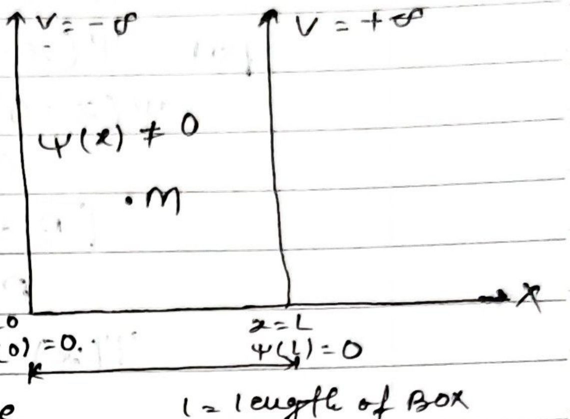
V.V.I

6 marks

Particle in a 1-Dimensional Box.

m = mass of particle which is inside the box of length 'L'



particle is free inside the box

potential energy $\boxed{V=0}$

ψ(x) → represent state of particle

$V = 0 \qquad 0 < x < L$

$V = +\infty \qquad x \geq L,\ x \leq 0$

The time-independent equation for a free particle in

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

Multiply both sides by $-\frac{2m}{\hbar^2}$

$$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi$$

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0$$

Put $\frac{2mE}{\hbar^2} = K^2$ ------ (1)

$$\frac{d^2\psi}{dx^2} + K^2\psi = 0 \quad \text{------ (2)}$$

The solution of eqⁿ (2) is

$$\psi(x) = A\sin kx + B\cos kx \quad \text{---- (3)}$$

Date :

(i) Boundary condition of $x = 0$

at $x = 0$, $\psi(x=0) = 0$

put $\psi(x=0) = 0$ and $x = 0$ in --- (3)

$$\psi(x=0) = A\sin 0 + B\cos 0$$

$$0 = 0 + B$$

$$\boxed{B = 0}$$

Put $B = 0$ in eqn (3)

$$\psi(x) = A\sin kx \quad \text{----} (4)$$

(ii) Second Boundary condition at $x = L$, $\psi(x=L) = 0$

put this value in (4)

$$\psi(x=L) = 0 = A\sin kL$$

$$= A\sin kL = 0 = \sin n\pi$$

$\therefore n = 1, 2, 3, 4 ---$

$$\Rightarrow kL = n\pi$$

$$\boxed{k = \frac{n\pi}{L}} \rightarrow \text{put this value in (1)}$$

$$\frac{2mE}{\hbar^2} = \frac{n^2\pi^2}{L^2}$$

$$\boxed{E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}}$$

$n = 1, 2, 3, 4 ----$

$n_1 = E_1 = \dfrac{\pi^2\hbar^2}{2mL^2}$ = ground state energy

$n_2 = E_2 = 4\underbrace{\dfrac{\pi^2\hbar^2}{2mL^2}}_{E_1} = 4E_1$ = Energy of first exicted state

Date :

$n_3 = E_3 = \frac{9\pi^2\hbar^2}{2mL^2} \rightarrow E_1$ $= 9E_1$ = second energy of excited state.

$n=4$ ——————— $16E_1$

$n=3$ ——————— $9E_1$

$n=2$ ——————— $4E_1$

$n=1$ ——————— E_1

To find wave function of particle

$$\psi(x) = A\sin kx$$

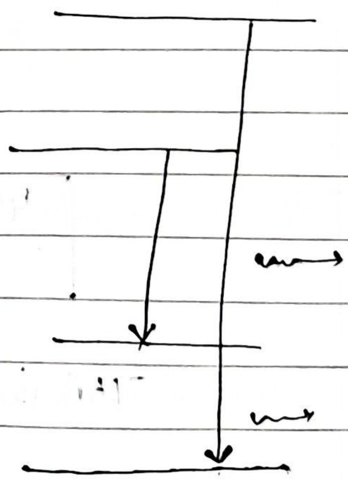
put $k = \frac{n\pi}{L}$

$$\psi(x) = A\sin\frac{n\pi x}{L} \quad \text{--- (5)}$$

Here, A = normalization constant.

The condition to normalize of wave function is

$$\int_{-\infty}^{+\infty} \psi^* \psi \, dx = 1$$



$$\psi^*(x) = A\sin\frac{n\pi x}{L}$$

$$\int_0^L A\sin\frac{n\pi x}{L}\, dx = 1$$

$$A^2\int_0^L \sin^2\frac{n\pi x}{L}\, dx = 1$$

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

Date :

$$\sin^2 \frac{n\pi x}{L} = \frac{1 - \cos \frac{2n\pi x}{L}}{2}$$

$$A^2 \int_0^L \frac{1 - \cos \frac{2n\pi x}{L}}{2} dx = 1$$

$$\frac{A^2}{2} \int_0^L \left(1 - \cos \frac{2n\pi x}{L}\right) dx = 1$$

$$\frac{A^2}{2} \left[x - \frac{\sin \frac{2n\pi x}{L}}{\frac{2n\pi}{L}}\right]_0^L = 1$$

$$\frac{A^2}{2} \left[(L-0) - \frac{L}{2n\pi} \left(\sin \frac{2n\pi . L}{L} - \sin 0\right)\right] = 1$$

$n = 1, 2, 3 \ldots$

$$\frac{A^2}{2} L = 1$$

$$\Rightarrow A^2 = \frac{2}{L}$$

$$A = \sqrt{\frac{L}{2}}$$

Put in eq (5)

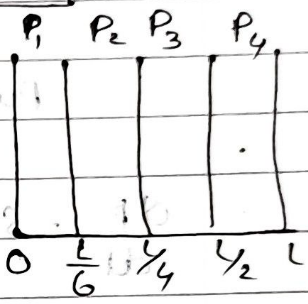
$$\boxed{\psi(x) = \sqrt{\frac{L}{2}} \sin \frac{n\pi x}{L}}$$

This is wave function of particle in 1-D Box.

Probability Current:

$$S = \frac{dP}{dt} = \text{probability current}$$

To prove :- $\boxed{S = \frac{-i\hbar}{2m}\left(\psi^* \nabla \psi - \psi \nabla \psi^*\right)}$



The probability of finding the particle in a region is given by

$$\boxed{P = \int \psi^* \psi \, dx} \quad \text{---} (1)$$

1 marks

Take derivative of (1) both sides. w.r.t 't'

$$\frac{dP}{dt} = \frac{d}{dt}\int \psi^* \psi \, dx$$

$$\frac{dP}{dt} = \int \frac{d}{dt}\left(\psi^* \psi\right) dx$$

ψ^* = complex consuget of ψ

$$\frac{dP}{dt} = \int \left(\psi^* \frac{d\psi}{dt} + \frac{d\psi^*}{dt}\psi\right) dx$$

$$\frac{dP}{dt} = \int \psi^* \frac{d\psi}{dt} dx + \int \psi \frac{d\psi^*}{dt} dx \quad \text{-----} (2)$$

Now the time-dependent schrodinger eqn

$$\left[\, i\hbar \frac{d\psi}{dt} = \frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi \,\right] \rightarrow \text{Remember}$$

$$\int f(x)\,g(x) = f(x)\int g(x)\,dx - \int \frac{df(x)}{dx}\int g(x)\,dx$$

Date :

$$\frac{d\psi}{dt} = \frac{1}{i\hbar}\left[-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi\right] \quad \text{--- (3)}$$

Take complex consugate both sides

$$\frac{d\psi^*}{dt} = \frac{1}{-i\hbar}\left[\frac{-\hbar^2}{2m}\frac{d^2\psi^*}{dx^2} + V\psi^*\right] \quad \text{--- (4)}$$

Put the value (3) & (4) in eq (2)

$$\frac{dP}{dt} = S = \int \psi^* \frac{1}{i\hbar}\left(\frac{-\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi\right)dx + \frac{\psi}{-i\hbar}\left(\frac{-\hbar^2}{2m}\frac{d^2\psi^*}{dx^2} + V\psi^*\right)dx$$

$$= \int\left[\frac{-\hbar}{i2m}\,\psi^*\frac{d^2\psi}{dx^2} + \cancel{\frac{\psi^*}{i\hbar}V\psi} + \frac{\hbar}{2mi}\,\psi\frac{d^2\psi^*}{dx^2} - \cancel{\frac{1}{i\hbar}\psi V\psi^*}\right]dx$$

$$= \frac{-\hbar}{2mi}\int\left(\psi^*\frac{d^2\psi}{dx^2} - \psi\frac{d^2\psi^*}{dx^2}\right)dx$$

$$= \frac{-\hbar}{2mi}\left\{\left[\psi^*\frac{d\psi}{dx} - \int\frac{d\psi^*}{dx}\frac{d\psi}{dx}dx\right] - \left[\psi\frac{d\psi^*}{dx} - \int\frac{d\psi}{dx}\frac{d\psi^*}{dx}dx\right]\right\}$$

$$= \left[\psi^*\frac{d\psi}{dt} - \int\cancel{\frac{d\psi^*}{dx}\frac{d\psi}{dx}}dx - \psi\frac{d\psi^*}{dt} + \int\cancel{\frac{d\psi}{dx}\frac{d\psi^*}{dx}}\right]$$

$$\boxed{S = \frac{dP}{dt} = \frac{i\hbar}{2m}\left[\psi^*\frac{d\psi}{dx} - \psi\frac{d\psi^*}{dx}\right]} \rightarrow \text{Probability Current.}$$

Date :

wave packet :

→ wave packet is a localized wave created by superposition of multiple of different frequency.

① Phase velocity $V_p = \frac{\omega}{k}$

→ Phase velocity is the velocity at which a single point of wave propagates.

②. Particle velocity

→ It is a velocity of every particle in the medium ossicilates due to wave.

③ Group velocity $V_g = \frac{d\omega}{dk}$

ω = Angular frequency

k = wave vector

$$\boxed{k = \frac{2\pi}{\lambda}}$$

→ The velocity with reach the overall the shape or envelope of wave packet propagates.

Date :

N.V.S

Relation Between Phase velocity and Group velocity

Phase velocity $V_p = \frac{\omega}{k} \Rightarrow \omega = kV_p$

Group velocity $V_g = \frac{d\omega}{dk}$

$V_g = \frac{d\omega}{dk}$

$V_g = \frac{d}{dk}(kV_p)$

$V_g = k\frac{dV_p}{dk} + \frac{dk}{dk}V_p$

$V_g = V_p + k\frac{dV_p}{dk}$

$V_g = V_p + k\frac{dV_p}{d\lambda}\cdot\frac{d\lambda}{dk}$ — (1)

$k = \frac{2\pi}{\lambda}$

$\frac{dk}{d\lambda} = \frac{-2\pi}{\lambda^2}$

$\frac{d\lambda}{dk} = \frac{-\lambda^2}{2\pi}$

put $\frac{d\lambda}{dk} = \frac{-\lambda^2}{2\pi}$ in R.H.S (1)

$V_g = V_p + \frac{2\pi}{\lambda}\cdot\left(\frac{-\lambda^2}{2\pi}\right)\frac{dV_p}{d\lambda}$

$$\boxed{V_g = V_p - \lambda\frac{dV_p}{d\lambda}}$$

(1) mark

Que. Define (a) phase velocity (b) Group velocity (c) Particle velocity.

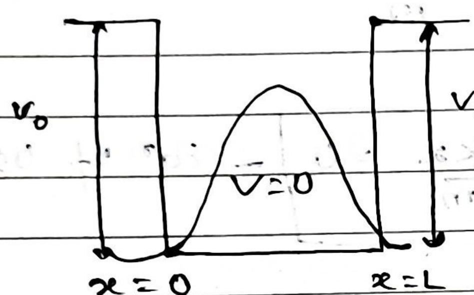
finite Potential value well.

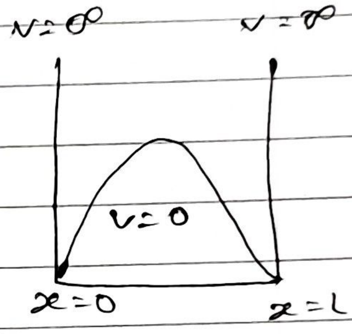
$V_0 = 0 \quad 0 < x < L$ $\qquad\qquad V = 0 \quad x \le x \le L$

$x > 0$

$V = V_0 \quad x \le 0$ $\qquad\qquad V = +\infty \quad x \le 0$

$x \ge 0$



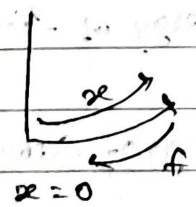


Turneling in finite Potential well

1 marks

Turneling is a quantum mechanical phenomenon where particles has non-zero probability of penetrating a potential energy barrier. if its total energy is less than the ~~wat~~ barrier energy.

Harmonic Oscillator



Restoring force $\quad f \propto x$

$$f = -kx$$

K = force constant

Date :

By Newton's law

$$F = ma$$

$$ma = -kx$$

$$m\frac{d^2x}{dt^2} = -kx$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\boxed{\frac{d^2x}{dt^2} + \frac{k}{m}x = 0}$$ — Eqn of Oscillator.

Put $\omega^2 = \frac{k}{m} \Rightarrow \boxed{k = m\omega^2}$

$$\boxed{\frac{d^2x}{dt^2} + \omega^2 x = 0}$$

marks Energy of Quantum Harmonic Oscillator

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega}$$

$n = 0, 1, 2, 3 \ldots$

$n = 0$: $E_0 = \frac{1}{2}\hbar\omega$ = ground state velocit

$n = 1$ $E_1 = \frac{3}{2}\hbar\omega$ = Energy of first excited state

$n = 2$ $E_2 = \frac{7}{2}\hbar\omega$ = Energy of Second Excited state